

# Phase 1: Collaborative Environment & Foundation Setup

## AI-Powered SOC & Smart Threat Detection System

| Item                | Details                                               |
|---------------------|-------------------------------------------------------|
| Project             | AI-Powered SOC & Smart Threat Detection System        |
| Phase               | Phase 1: Collaborative Environment & Foundation Setup |
| Team Size           | 10 members                                            |
| Simulation Platform | PNetLab 4.2.10 on VMware Workstation                  |
| Remote Access       | Radmin VPN (Graduation_Lab)                           |

### 1. Introduction to the Workspace

A centralized and flexible workspace was set up so that the project team of 10 members can work remotely and simultaneously on building the infrastructure of the "AI-Powered SOC & Smart Threat Detection System" project. The PNetLab network emulator is hosted on a central Host Server running VMware Workstation, which consolidates resources and avoids version conflicts between team members.

The following table lists the specifications allocated to the virtual server that hosts PNetLab:

Table 1: Virtual Server (PNetLab VM) Specifications

| Component              | Specification                                                 |
|------------------------|---------------------------------------------------------------|
| PNetLab Version        | 4.2.10                                                        |
| Guest Operating System | Ubuntu 18.04.5 LTS                                            |
| Hypervisor             | VMware Workstation (Hardware compatibility: Workstation 10.x) |
| Memory (RAM)           | 16 GB+                                                        |
| Processors             | 8                                                             |
| Hard Disk (SCSI)       | 100 GB                                                        |
| Network Adapters       | 2 (Adapter 1: VMnet0 Bridged / Adapter 2: VMnet8 NAT)         |

### 2. VPN & Peer-to-Peer (P2P) Overlay Network

To connect the team's devices to the central server securely, without exposing any service to the Public Internet, Radmin VPN was used.

- **Technology:** The software relies on Peer-to-Peer (P2P) technology to create a Virtual LAN that works as an Overlay Network, so the operating system treats the connected devices as if they were attached to a single local Switch.

The following table summarizes the VPN network settings adopted in the project:

Table 2: Radmin VPN Network Settings

| Parameter             | Value                                                      |
|-----------------------|------------------------------------------------------------|
| Network Name          | Graduation_Lab                                             |
| VPN Subnet (IP Range) | 26.0.0.0/8                                                 |
| Host Server VPN IP    | 26.10.10.101                                               |
| Web Access Endpoint   | http://26.249.209.213 (forwarded to 26.10.10.101, port 80) |

### Web Interface Access via Port Forwarding

To make the PNetLab web interface reachable by every team member over the VPN, an HTTP port-forwarding rule was created using the Windows netsh utility. The rule listens on the Radmin address 26.249.209.213 and relays the traffic to the PNetLab server, with the following parameters:

- **Listen address / port:** 26.249.209.213 / 80
- **Connect address / port:** 26.10.10.101 / 80 (PNetLab server)

```
C:\Users\PC> netsh interface portproxy add v4tov4 listenport=80  
listenaddress=26.249.209.213 connectport=80 connectaddress=26.10.10.101
```

Figure 1: netsh portproxy command forwarding HTTP (port 80) from 26.249.209.213 to the PNetLab server 26.10.10.101

After the rule was applied, the PNetLab web interface became accessible from the browser at <http://26.249.209.213>. The Running Lab Sessions page shows the shared Graduation\_Project lab running with 4 active nodes, which confirms that the collaborative environment works end to end.

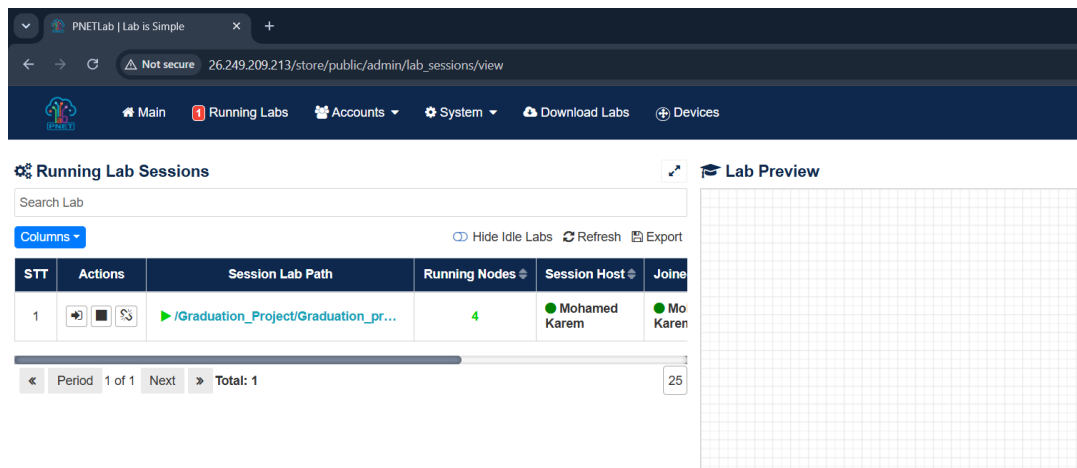


Figure 2: PNetLab web interface accessed through the forwarded address, showing the running Graduation\_Project lab session

### 3. Hypervisor Network Configuration (VMware Workstation)

To ensure security isolation and provide separate paths for management and Internet traffic, two Network Adapters were assigned to the virtual server through the Virtual Network Editor settings:

#### Network Adapter 1 (Management & Collaboration):

- **Type:** Connected to VMnet0 and set to Bridged mode.
- **Routing:** The Bridge was directed to the virtual adapter Famatech Radmin VPN Ethernet Adapter.

- **Purpose:** Allow team members to reach the PNetLab graphical management interface (GUI) through a web browser using the VPN address.

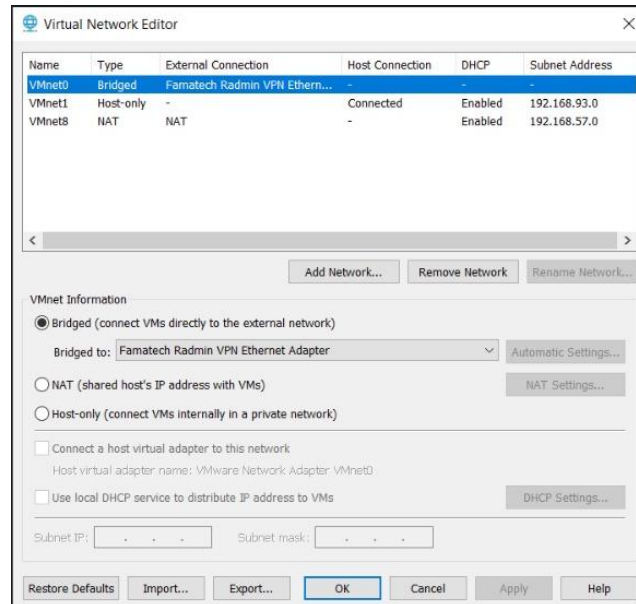


Figure 3: VMnet0 configured in Bridged mode and bound to the Famatech Radmin VPN Ethernet Adapter

### Network Adapter 2 (Internet Access & External Threats):

- **Type:** Connected to VMnet8 and set to Network Address Translation (NAT) mode.
- **Configuration:** The Use local DHCP service option was enabled to distribute addresses automatically (Dynamic Host Configuration Protocol) to the devices inside the simulation.
- **Purpose:** Provide real Internet connectivity for servers (such as Ubuntu) to download packages and updates, and build an external network (WAN) to simulate external attacks.
- **NAT Subnet:** 192.168.57.0/24 (Subnet Mask: 255.255.255.0)

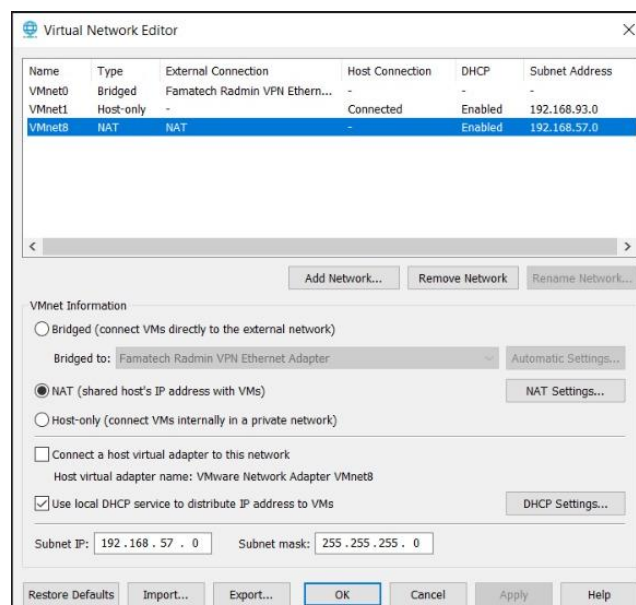


Figure 4: VMnet8 configured in NAT mode with the DHCP service enabled on the 192.168.57.0/24 network

The following table summarizes the settings of the two network adapters of the virtual server:

Table 3: Summary of Virtual Network Adapter Settings

| Adapter          | VMnet / Mode     | Connected To                         | Addressing                          | Purpose                            |
|------------------|------------------|--------------------------------------|-------------------------------------|------------------------------------|
| <b>Adapter 1</b> | VMnet0 / Bridged | Famatech Radmin VPN Ethernet Adapter | 26.0.0.0/8 (server IP 26.10.10.101) | Management & collaboration         |
| <b>Adapter 2</b> | VMnet8 / NAT     | NAT service of the host machine      | 192.168.57.0/24 (DHCP enabled)      | Internet access & external threats |

## Verification of Network Interfaces

After the server was started, the PNetLab terminal was used to verify the addresses assigned to its interfaces. The following was observed:

- **pnet0**: Received the address 26.10.10.101, the same address as the server on the Radmin VPN network, which makes the management interface available to team members.
- **pnet\_nat**: Internal interface used by PNetLab, with the address 10.0.137.1.
- **docker0**: Internal interface of the Docker service, with the address 10.177.0.1.
- The system message also confirms that the service is working normally (Your service is working normally).

```
PNetLab (default root password is 'pnet')
Use https or http://26.10.10.101/

pnetlab login: Your service is working normally

PNetLab (default root password is 'pnet')
Use https or http://26.10.10.101/

pnetlab login: root
Password:
Last login: Sat Oct 3 21:34:15 UTC 2026 on tty1
Welcome to Ubuntu 18.04.5 LTS

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/advantage

System information as of Sun Oct 4 02:04:05 UTC 2026

System load: 1.09          Users logged in: 0
Usage of /: 8.9% of 96.94GB IP address for pnet0: 26.10.10.101
Memory usage: 19%         IP address for pnet_nat: 10.0.137.1
Swap usage: 0%            IP address for docker0: 10.177.0.1
Processes: 213

* Canonical Livepatch is available for installation.
- Reduce system reboots and improve kernel security. Activate at:
  https://ubuntu.com/livepatch

root@pnetlab:~# _
```

Figure 5: PNetLab terminal showing the pnet0 interface address (26.10.10.101) and a healthy service status

## 4. PNetLab Topology & External Connectivity (WAN)

Inside the PNetLab environment, the VMware network adapters were converted into Clouds to connect the virtual devices to the outside world:

- **Management Network:** Represented by Management(Cloud0) and bound to the management adapter (Radmin).
- **WAN / Internet Network:** Represented by Cloud1 and bound to the NAT adapter.
- **Troubleshooting:** To avoid the DHCP Drop Bug, a common issue when a Cloud is connected directly to Routers/Firewalls, an Unmanaged Switch was added to act as an intermediate distribution point between Cloud1 and the devices.
- **Implementation:** The WAN Port of the FortiGate firewall and the external attacker system (Kali Linux) will both be connected to this switch, so that they obtain IP addresses in the same external network, creating a realistic environment for external intrusion scenarios.

The following diagram shows the two separate paths for management and Internet traffic, from the team members down to the devices inside the emulator:

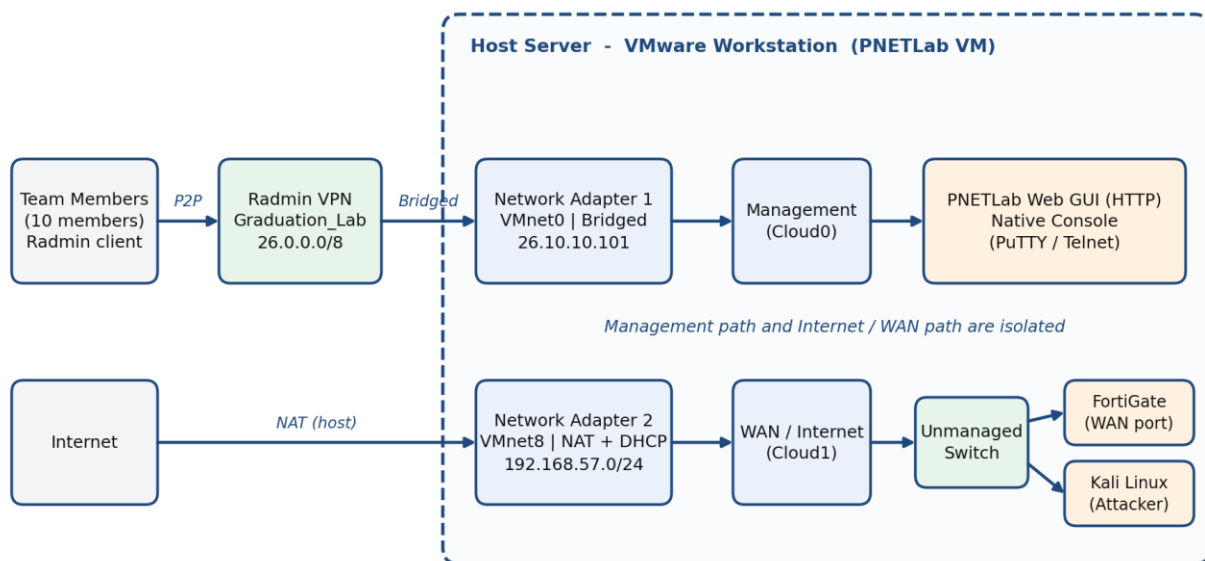


Figure 6: Logical diagram linking the PNetLab environment to the management network (Radmin VPN) and the Internet/WAN network (NAT)

## 5. Collaborative Workflow & Terminal Management

To avoid synchronization and isolation problems (Pod Isolation / Parallel Versions) caused by the browser's built-in HTML5 console when several members work at the same time:

- The PNetLab Windows Client Pack was installed on the team members' devices.
- The Native Console option was enabled to rely on PuTTY as the primary Terminal, connecting directly to the Ports of the virtual devices over the Telnet protocol.
- This resolved the Connection Refused disconnections and ensured a unified, immediate view of all Configurations made by any team member.

## 6. Summary of Phase 1 Outcomes

This phase produced a ready collaborative workspace made up of the following components:

*Table 4: Summary of Core Infrastructure Components*

| Component                          | Role                                                                                             |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>VMware Workstation</b>          | Hosts the central PNetLab server and provides the virtual network adapters                       |
| <b>Radmin VPN</b>                  | Connects team members to the server through a secure Overlay network (26.0.0.0/8)                |
| <b>Port Forwarding (netsh)</b>     | Publishes the PNetLab web interface to team members over the VPN (26.249.209.213 → 26.10.10.101) |
| <b>VMnet0 (Bridged)</b>            | Management and collaboration path, and access to the PNetLab interface                           |
| <b>VMnet8 (NAT + DHCP)</b>         | Internet and WAN path (192.168.57.0/24)                                                          |
| <b>Unmanaged Switch</b>            | Intermediate distribution point that avoids the DHCP Drop Bug                                    |
| <b>PNetLab Client Pack + PuTTY</b> | Simultaneous team work through the Native Console without disconnections                         |